Oren Segal

Applied AI Engineer · Agents, Evals and LLM Reliability

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SUMMARY

Applied AI engineer who builds agents and the evals that show whether they work. Built an open-source incident agent, **rootcause**, and its eval: 525 runs through Claude Code's agent loop across seven configurations of model, effort and retrieval: accuracy held at **77-84%** while cost per solved incident varied **25x**. Building Shelfie, a vision-based iOS app in closed beta, with Claude Code agents under my architecture, review and CI. Before that, five years as a senior data analyst at Reshet 13, an Israeli TV network, building the pricing and forecasting models the business ran on.

EXPERIENCE

Independent · Applied AI Engineer, Shelfie and open-source agent tooling — New York · 2025 - Present

Shelfie · iOS kitchen-inventory app, closed TestFlight beta

- Architected and built the app end to end: Swift 6/SwiftUI with Watch, widget and iMessage extensions, on a Supabase backend (Postgres, auth, storage, about 60 TypeScript edge functions).
- Built a vision pipeline for shaky, badly lit kitchen footage: drops blurry frames, prompts for light, resolves food on device (Apple Vision, OCR, barcodes) and sends only unresolved items to a cloud model.
- Routed every model call through one LLM gateway with spending limits, caching and a circuit breaker, enforced by a CI rule; replaced an agent's scan-cost estimate that was **about 34x too low** with measured billing.
- Direct the Claude Code agents that write most of the code; own the architecture, review and CI that fails zero-test runs, flags weakened tests and holds merges for human review.

Career transition · Relocated to the US, full-time AI/ML retraining — New York · 2023 - 2025

Reshet 13 · Senior Data Analyst — Tel Aviv · 2018 - 2023

- Built a regression model that found underpriced primetime ad slots, then worked with the commercial team until the new prices went live; **revenue rose 17%** in FY2022.
- Developed the network's main viewership forecasting model, the standard input for leadership's content acquisition and greenlight decisions.
- Replaced 70+ weekly Excel reports with Python/SQL pipelines, saving **about 70 hours a week**; mentored four junior analysts.

Cohort NYC · Volunteer Board Member — New York · 2026 - Present

- Built a Postgres pipeline tying each Luma registration to the link that brought it in, showing which channels worked; subscriptions and attendance **doubled in 75 days** (1,400+ subscribers).

OPEN-SOURCE PROJECTS

rootcause · github.com/OrenSegal/rootcause · other work at github.com/OrenSegal

- **rootcause:** incident and CI root-cause agent with its own tool loop over MCP (no framework), an approval gate for production actions, crash-safe resume and OpenTelemetry traces. Ships as a CLI, MCP server and GitHub Action.
- Designed its eval: 25 incidents with known causes and planted decoys, repeated runs, Wilson intervals, pass^k, cost per solved incident. A **300-run** test found multi-agent orchestration tripled cost for no measurable gain.
- Benchmarked retrieval over 33 postmortems: BM25 reached recall@5 of **0.94** versus 0.86 for a transformer embedding model running about 1,000x slower, so the agent ships with lexical search.
- **sous**, the harness my agents run in, rejects test logs where nothing ran and catches tests deleted or skipped to reach green. **litmus** mutation-tests eval suites and lists the broken copies they miss.

SKILLS

AI engineering	agents and tool use, multi-agent orchestration (hand-written and LangGraph), MCP, RAG (BM25, embeddings, hybrid), LLM evals, LLM-as-judge calibration, LLM gateways, OpenTelemetry
Models and tools	Claude API, Claude Code and subagents, Gemini, Apple Vision and Foundation Models, pgvector
Engineering	Python, TypeScript, Swift/SwiftUI, SQL, PostgreSQL, Supabase, Docker, Azure, GitHub Actions, pytest
Data science	regression, forecasting, confidence intervals, experiment and eval design

EDUCATION & CERTIFICATES

B.Sc. Information Systems, The Academic College of Tel Aviv-Yaffo — 2018

Certificates (2025): MLOps & LLMOps Specializations, Duke · Agent Engineering Bootcamp, Break into Data